Ejercicios Resueltos Sobre Transformadas de Fourier

1) Hallar la Transformada de Fourier de las siguientes funciones.

   a) $x(t) = e^{-a\,t} \quad t \geq 0 \quad \wedge \quad a > 0$

   b) $f(t) = u\left(t + \dfrac{a}{2}\right) - u\left(t - \dfrac{a}{2}\right)$

   c) $\Lambda_n : [-n, n] \to \mathbb{R} : \Lambda_n(t) = \dfrac{n - |t|}{n^2}$

   d) $f(t) = e^{\beta\,t}\,e^{i\,\alpha\,t} \quad t \geq 0$

   e) $f(t) = e^{-t^2}$

Resolución:

a) $\mathcal{F}\{x(t)\} = \displaystyle\int_0^{\infty} e^{-a\,t}\,e^{-i\,\omega\,t}\,dt \implies \mathcal{F}\{x(t)\} = \int_0^{\infty} e^{-(a+i\,\omega)\,t}\,dt$. Ahora $\displaystyle\lim_{\lambda\to\infty}\int_0^{\lambda} e^{-(a+i\,\omega)\,t}\,dt = \lim_{\lambda\to\infty} \left.\frac{e^{-(a+i\,\omega)\,t}}{-(a+i\,\omega)}\right|_0^{\lambda}$

$$\lim_{\lambda\to\infty}\frac{e^{-(a+i\,\omega)\,\lambda} - 1}{-(a+i\,\omega)} = \frac{1}{a+i\,\omega} \implies F(\omega) = \frac{1}{a+i\,\omega}$$

b) $\mathcal{F}\{f(t)\} = \displaystyle\int_{-a/2}^{a/2} e^{-i\,\omega\,t}\,dt = \left.\frac{e^{-i\,\omega\,t}}{-i\,\omega}\right|_{-a/2}^{a/2} = \frac{e^{i\,\omega\,\frac{a}{2}} - e^{-i\,\omega\,\frac{a}{2}}}{i\,\omega} \implies F(\omega) = 2\frac{sen\left(\omega\,\frac{a}{2}\right)}{\omega} \implies F(\omega) = a\frac{sen\left(\omega\,\frac{a}{2}\right)}{\omega\,\frac{a}{2}}$

c) $\mathcal{F}\{\Lambda_n(t)\} = \displaystyle\int_{-n}^{n}, \frac{n - |t|}{n^2}\,e^{-i\,\omega\,t}\,dt = dt$

$$= \frac{1}{n}\int_{-n}^{n} e^{-i\,\omega\,t}\,dt - \frac{1}{n^2}\int_{-n}^{n} |t|\,e^{-i\,\omega\,t}\,dt = \left.\frac{e^{-i\,\omega\,t}}{-i\,\omega\,n}\right|_{-n}^{n} - \frac{1}{n^2}\left[-\int_{-n}^{0} t\,e^{-i\,\omega\,t}dt + \int_0^{n} t\,e^{-i\,\omega\,t}dt\right]$$

Ahora si en la primera integral se hace $t = -t \implies dt = -dt$, se tiene $\displaystyle\int_n^0 t\,e^{i\,\omega\,t}dt = \int_0^n t\,e^{i\,\omega\,t}dt$ Reemplazando se tiene

$$\mathcal{F}\{\Lambda_n(t)\} = \frac{e^{i\,\omega\,n} - e^{-i\,\omega\,n}}{i\,\omega\,n} - \frac{1}{n^2}\int_0^{n} t\,(e^{i\,\omega\,t} + e^{-i\,\omega\,t})\,dt$$

$$\mathcal{F}\{\Lambda_n(t)\} = 2\,\frac{sen(\omega\,n)}{\omega\,n} - \frac{2}{n^2}\int_0^{n} t\,cos(\omega\,t)\,dt = -\frac{2}{n^2}\left.\left[\frac{cos(\omega\,t)}{\omega^2} + \frac{t\,sen(\omega\,t)}{\omega}\right]\right|_0^{n}$$

$$\mathcal{F}\{\Lambda_n(t)\} = -\frac{2}{n^2}\int_0^{n} t\,cos(\omega\,t)\,dt = 2\,\frac{sen(\omega\,n)}{\omega\,n} - \frac{2}{n^2}\left[\frac{cos(\omega\,n)}{\omega^2} + \frac{n\,sen(\omega\,n)}{\omega} - \frac{1}{\omega^2}\right]$$

$$\mathcal{F}\{\Lambda_n(t)\} = -2\,\frac{cos(\omega\,n)}{(\omega\,n)^2} + \frac{2}{(\omega\,n)^2}$$

$$\mathcal{F}\{\Lambda_n(t)\} = 2\,\frac{1 - cos(\omega\,n)}{(\omega\,n)^2} = \frac{sen^2\left(\frac{\omega\,n}{2}\right)}{\left(\frac{\omega\,n}{2}\right)^2} \implies F(\omega) = sinc^2\left(\frac{\omega\,n}{2}\right)$$

d) $\mathcal{F}\{f(t)\} = \displaystyle\int_0^{\infty} e^{(\beta+i\,\alpha)\,t}\,e^{-i\,\omega\,t}\,dt = \int_0^{\infty} e^{(\beta+i\,(\alpha-\omega))\,t}\,dt$